

REPORT DOCUMENT

- TASK 1:

```
file 1.cpp  Untitled2
1  #include<iostream>
2  using namespace std;
3  struct Country {
4      string name;
5      float population;
6  };
7  int main() {
8      Country c;
9      cout << "Enter country name: ";
10     cin >> c.name;
11     cout << "Enter population (in millions): ";
12     cin >> c.population;
13     cout << "\nCountry Name: " << c.name << endl;
14     cout << "Population: " << c.population << " million" << endl;
15     return 0;
16 }
```

```
C:\Users\MSi\Desktop\file 1.e:  ×  +  v
Enter country name: pakistan
Enter population (in millions): 12000000

Country Name: pakistan
Population: 1.2e+007 million

-----
Process exited after 12.03 seconds with return value 0
Press any key to continue . . . |
```

- TASK 2:

```
1 #include<iostream>
2 using namespace std;
3 struct Student {
4     int rollNo;
5     float marks[5];
6 };
7 int main() {
8     Student s;
9     cout << "Enter Roll No: ";
10    cin >> s.rollNo;
11    for(int i = 0; i < 5; i++) {
12        cout << "Enter marks for subject " << i+1 << ": ";
13        cin >> s.marks[i];
14    }
15    cout << "\nRoll No: " << s.rollNo << endl;
16    for(int i = 0; i < 5; i++) {
17        cout << "Subject " << i+1 << " Marks: " << s.marks[i] << endl;
18    }
19    return 0;
20 }
```

```
C:\Users\MSi\Desktop\file 2.e x + v
Enter Roll No: 5
Enter marks for subject 1: 50
Enter marks for subject 2: 78
Enter marks for subject 3: 69
Enter marks for subject 4: 70
Enter marks for subject 5: 69

Roll No: 5
Subject 1 Marks: 50
Subject 2 Marks: 78
Subject 3 Marks: 69
Subject 4 Marks: 70
Subject 5 Marks: 69

-----
Process exited after 28.23 seconds with return value 0
Press any key to continue . . . |
```

- TASK 3:

file 3.cpp

```
1  #include<iostream>
2  using namespace std;
3  struct Car {
4      string model;
5      string color;
6      float price;
7  };
8  int main() {
9      Car cars[5];
10     for(int i = 0; i < 5; i++) {
11         cout << "Enter model of car " << i+1 << ": ";
12         cin >> cars[i].model;
13         cout << "Enter color of car " << i+1 << ": ";
14         cin >> cars[i].color;
15         cout << "Enter price of car " << i+1 << ": ";
16         cin >> cars[i].price;
17     }
18     int choice;
19     cout << "\nEnter 1 to see most costly car, 2 to see all cars: ";
20     cin >> choice;
21     if(choice == 1) {
22         int maxIndex = 0;
23         for(int i = 1; i < 5; i++) {
24             if(cars[i].price > cars[maxIndex].price)
25                 maxIndex = i;
26         }
27         cout << "\nMost Costly Car:" << endl;
28         cout << "Model: " << cars[maxIndex].model << endl;
29         cout << "Color: " << cars[maxIndex].color << endl;
30         cout << "Price: " << cars[maxIndex].price << endl;
31     }
32     else if(choice == 2) {
33         for(int i = 0; i < 5; i++) {
34             cout << "\nCar " << i+1 << ":" << endl;
35             cout << "Model: " << cars[i].model << endl;
36             cout << "Color: " << cars[i].color << endl;
37             cout << "Price: " << cars[i].price << endl;
38         }
39     }
40     return 0;
41 }
```

```
it Search View Project Execute Tools AStyle Window Help
C:\Users\MSI\Desktop\file 3.e × + ∨

Enter model of car 1: 2020
Enter color of car 1: red
Enter price of car 1: 3200000
Enter model of car 2: 2021
Enter color of car 2: white
Enter price of car 2: 3500000
Enter model of car 3: 2022
Enter color of car 3: black
Enter price of car 3: 4000000
Enter model of car 4: 2023
Enter color of car 4: blue
Enter price of car 4: 4500000
Enter model of car 5: 2024
Enter color of car 5: grey
Enter price of car 5: 4700000

Enter 1 to see most costly car, 2 to see all cars: 1

Most Costly Car:
Model: 2024
Color: grey
Price: 4.7e+006
```

- TASK 4:

```
1 #include<iostream>
2 using namespace std;
3
4 struct Phone {
5     int areaCode;
6     int exchange;
7     int number;
8 };
9 int main() {
10     Phone p1 = {212, 767, 8900};
11     Phone p2;
12     cout << "Enter your area code, exchange, and number: ";
13     cin >> p2.areaCode >> p2.exchange >> p2.number;
14     cout << "My number is (" << p1.areaCode << ") "
15         << p1.exchange << "-" << p1.number << endl;
16     cout << "Your number is (" << p2.areaCode << ") "
17         << p2.exchange << "-" << p2.number << endl;
18     return 0;
19 }
```

```
C:\Users\MSi\Desktop\task 4.  X  +  v

Enter your area code, exchange, and number: +92
30
3211
My number is (212) 767-8900
Your number is (92) 30-3211

-----
Process exited after 32.36 seconds with return value 0
Press any key to continue . . . |
```

- **TASK 5;**

```
file 5.cpp

1  #include<iostream>
2  using namespace std;
3  struct Point {
4      int x;
5      int y;
6  };
7  int main() {
8      Point p1, p2, p3;
9
10     cout << "Enter coordinates for p1: ";
11     cin >> p1.x >> p1.y;
12     cout << "Enter coordinates for p2: ";
13     cin >> p2.x >> p2.y;
14     p3.x = p1.x + p2.x;
15     p3.y = p1.y + p2.y;
16     cout << "Coordinates of p1+p2 are: " << p3.x << ", " << p3.y << endl;
17     return 0;
18 }
```

```
C:\Users\MSI\Desktop\file 5 n  X + v
ect Enter coordinates for p1: 2
3
Enter coordinates for p2: 3
4
Coordinates of p1+p2 are: 5, 7

-----
Process exited after 7.949 seconds with return value 0
Press any key to continue . . . |
```

- **TASK 6:**

```
file 6 new.cpp
1  #include<iostream>
2  using namespace std;
3  struct Distance {
4      int feet;
5      float inches;
6  };
7
8  struct Volume {
9      Distance length;
10     Distance width;
11     Distance height;
12 };
13 float Feet(Distance d) {
14     return d.feet + d.inches / 12.0;
15 }
16 int main() {
17     Volume room = {{12, 6}, {10, 4}, {8, 0}};
18     float l = Feet(room.length);
19     float w = Feet(room.width);
20     float h = Feet(room.height);
21     float volume = l * w * h;
22     cout << "Length : " << l << " ft" << endl;
23     cout << "Width : " << w << " ft" << endl;
24     cout << "Height : " << h << " ft" << endl;
25     cout << "Volume : " << volume << " cubic feet" << endl;
26     return 0;
27 }
```

```
using namespace std;
struct Distance {
    int feet;
};

Length : 12.5 ft
Width : 10.3333 ft
Height : 8 ft
Volume : 1033.33 cubic feet

-----
Process exited after 0.1005 seconds with return value 0
Press any key to continue . . . |
```

- TASK 7:

```
file 7 new.cpp
1  #include<iostream>
2  using namespace std;
3  struct Employee {
4      int empNumber;
5      float compensation;
6  };
7  int main() {
8      Employee emp[3];
9      for(int i = 0; i < 3; i++) {
10         cout << "Enter employee number for employee " << i+1 << ": ";
11         cin >> emp[i].empNumber;
12         cout << "Enter compensation for employee " << i+1 << ": $";
13         cin >> emp[i].compensation;
14     }
15     cout << "\nEmployee Records:" << endl;
16     for(int i = 0; i < 3; i++) {
17         cout << "\nEmployee " << i+1 << ":" << endl;
18         cout << "Number: " << emp[i].empNumber << endl;
19         cout << "Compensation: $" << emp[i].compensation << endl;
20     }
21     return 0;
22 }
```

```
C:\Users\MSI\Desktop\file 7 n x + v
Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9

Enter employee number for employee 1: 3
Enter compensation for employee 1: $50
Enter employee number for employee 2: 4
Enter compensation for employee 2: $60
Enter employee number for employee 3: 5
Enter compensation for employee 3: $70

Employee Records:

Employee 1:
Number: 3
Compensation: $50

Employee 2:
Number: 4
Compensation: $60

Employee 3:
Number: 5
Compensation: $70

-----
Process exited after 26.9 seconds with return value 0
Press any key to continue . . . |
```

- TASK 8:

```
file 8.cpp
1  #include<iostream>
2  using namespace std;
3  struct Date {
4      int month;
5      int day;
6      int year;
7  };
8  int main() {
9      Date d;
10     char slash;
11     cout << "Enter date (MM/DD/YYYY): ";
12     cin >> d.month >> slash >> d.day >> slash >> d.year;
13     cout << "Date: " << d.month << "/"
14         << d.day << "/" << d.year << endl;
15     return 0;
16 }
```



```
C:\Users\MSI\Desktop\file 8.e  X + v
Enter date (MM/DD/YYYY): 2/23/2021
Date: 2/23/2021

-----
Process exited after 15.26 seconds with return value 0
Press any key to continue . . . |
```

- TASK 9:

```
file 9.cpp
1  #include<iostream>
2  using namespace std;
3  struct Time {
4      int hours;
5      int minutes;
6      int seconds;
7  };
8  int main() {
9      Time t1;
10     char colon;
11     cout << "Enter time (HH:MM:SS): ";
12     cin >> t1.hours >> colon >> t1.minutes >> colon >> t1.seconds;
13     long totalSecs = t1.hours * 3600 + t1.minutes * 60 + t1.seconds;
14     cout << "Total seconds: " << totalSecs << endl;
15     return 0;
16 }
```

```
(globals)
C:\Users\MSI\Desktop\file 9.e  X + v
Enter time (HH:MM:SS): 2:35:57
Total seconds: 9357

-----
Process exited after 16.3 seconds with return value 0
Press any key to continue . . . |
```

- TASK 10:

file 10.cpp

```
1  #include<iostream>
2  using namespace std;
3  struct Time {
4      int hours;
5      int minutes;
6      int seconds;
7  };
8  int main() {
9      Time t1, t2, t3;
10     char colon;
11     cout << "Enter first time (HH:MM:SS): ";
12     cin >> t1.hours >> colon >> t1.minutes >> colon >> t1.seconds;
13     cout << "Enter second time (HH:MM:SS): ";
14     cin >> t2.hours >> colon >> t2.minutes >> colon >> t2.seconds;
15     long totalSecs = (t1.hours * 3600 + t1.minutes * 60 + t1.seconds)
16                   + (t2.hours * 3600 + t2.minutes * 60 + t2.seconds);
17     t3.hours = totalSecs / 3600;
18     t3.minutes = (totalSecs % 3600) / 60;
19     t3.seconds = totalSecs % 60;
20     cout << "Result: " << t3.hours << ":"
21          << t3.minutes << ":" << t3.seconds << endl;
22     return 0;
23 }
```

C:\Users\MSi\Desktop\file 10. x + v

```
Enter first time (HH:MM:SS): 2:47:56
Enter second time (HH:MM:SS): 3:23:50
Result: 6:11:46
```

```
-----
Process exited after 22.77 seconds with return value 0
Press any key to continue . . . |
```

- HOME TASK 1:

home task 1.cpp

```
1  #include<iostream>
2  using namespace std;
3  struct Hotel {
4      string name;
5      int dishes;
6      float bill;
7  };
8  int main() {
9      Hotel h[3];
10     float totalBill = 0;
11     for(int i = 0; i < 3; i++) {
12         cout << "Enter hotel name: ";
13         cin >> h[i].name;
14         cout << "Enter number of dishes: ";
15         cin >> h[i].dishes;
16         cout << "Enter total bill: ";
17         cin >> h[i].bill;
18         totalBill += h[i].bill;
19     }
20     float tax = totalBill * 0.15;
21     cout << "\nAll Hotel Records:" << endl;
22     for(int i = 0; i < 3; i++) {
23         cout << "\nHotel : " << h[i].name << endl;
24         cout << "Dishes: " << h[i].dishes << endl;
25         cout << "Bill : $" << h[i].bill << endl;
26     }
27     cout << "\nTotal Bill (before tax): $" << totalBill << endl;
28     cout << "Tax (15%):          $" << tax << endl;
29     cout << "Total Bill (after tax):  $" << totalBill + tax << endl;
30     return 0;
31 }
```

```

AStyle Window Help
TDM-GCC 4.9.2 64-bit B

C:\Users\MSI\Desktop\home >
Enter number of dishes: 8
Enter total bill: 34000
Enter hotel name: Al-Kauser
Enter number of dishes: 7
Enter total bill: 30000
Enter hotel name: Al-Maroon
Enter number of dishes: 5
Enter total bill: 27000

All Hotel Records:

Hotel : Al-Madina
Dishes: 8
Bill : $34000

Hotel : Al-Kauser
Dishes: 7
Bill : $30000

Hotel : Al-Maroon
Dishes: 5
Bill : $27000

Total Bill (before tax): $91000
Tax (15%): $13650
Total Bill (after tax): $104650

```

- HOME TASK 2:

home task 2.cpp

```
1  #include<iostream>
2  using namespace std;
3  struct Person {
4      float income;
5      float taxRate;
6      float tax;
7  };
8  int main() {
9      Person p[5];
10
11     for(int i = 0; i < 5; i++) {
12         cout << "Enter income for person " << i+1 << ": ";
13         cin >> p[i].income;
14         cout << "Enter tax rate for person " << i+1 << " (%): ";
15         cin >> p[i].taxRate;
16         p[i].tax = p[i].income * (p[i].taxRate / 100);
17     }
18     cout << "\nTax Payable:" << endl;
19     for(int i = 0; i < 5; i++) {
20         cout << "Person " << i+1
21             << " | Income: $" << p[i].income
22             << " | Tax Rate: " << p[i].taxRate << "%"
23             << " | Tax: $" << p[i].tax << endl;
24     }
25     return 0;
26 }
```

(globals)

```
C:\Users\MSI\Desktop\home · × + ∨
Enter income for person 1: 34000
Enter tax rate for person 1 (%): 12
Enter income for person 2: 37000
Enter tax rate for person 2 (%): 13
Enter income for person 3: 40000
Enter tax rate for person 3 (%): 15
Enter income for person 4: 42000
Enter tax rate for person 4 (%): 16
Enter income for person 5: 45000
Enter tax rate for person 5 (%): 18
```

Tax Payable:

Person 1		Income: \$34000		Tax Rate: 12%		Tax: \$4080
Person 2		Income: \$37000		Tax Rate: 13%		Tax: \$4810
Person 3		Income: \$40000		Tax Rate: 15%		Tax: \$6000
Person 4		Income: \$42000		Tax Rate: 16%		Tax: \$6720
Person 5		Income: \$45000		Tax Rate: 18%		Tax: \$8100